Chapter 1 · Energy Causes Change

GUIDED NOTES

Name: _____ Homeroom: ○ 4A ○ 4B ○ 4C ○ 4D

BQ Where can we observe evidence of energy causing change?

My first thinking

A pencil gets shorter and sharper.

V1 VOCABULARY — ENERGY & CHANGE

Reader p. 1

energy — the ability to cause **change** .

change — to become **different** .

N1 ENERGY CAUSES CHANGE

Reader pp. 1–2

Energy has the ability to cause change.

When you see a change, you are seeing **evidence** of ener

ENERGY	→	CHANGE	→	EVIDENCE

N2 DIFFERENT FORMS

Reader p. 3

Write one example that shows a change for each form.

FORM / EXAMPLE OF A CHANGE	
motion	a ball rolls across the floor
sound	a drum makes a noise
light	a flashlight lights up the room
heat	a warm mug warms my hands
electrical energy	a fan spins when plugged in

Sample answer — student wording may vary.

V2 STORED ENERGY

Reader p. 4

stored energy — energy that has the ability to cause change at a **later** time.

N3 OBJECTS CAN STORE ENERGY

Reader pp. 4–5

STORED ENERGY → **RELEASE** → **CHANGE**

What change could happen later?

battery (in a flashlight)

the flashlight lights up

stretched rubber band

it snaps and makes a sound

raised object (apple)

it falls down

Objects also have stored energy because of their **position**.

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N4 RECOGNIZING EVIDENCE

Reader pp. 1–5

Draw a change you observed. Show what it looked like BEFORE and AFTER.

BEFORE	AFTER
Sample: a dull pencil, long, rounded tip	Sample: shorter pencil, sharp tip, shavings

What changed?

The pencil got shorter and sharper.

CHECK YOUR THINKING

Reader pp. 1–5

How do we know energy is involved if we cannot see energy itself?

Start with: “We know because we see …”

We know because we see a change, like a ball starting to move

or a light turning on.

BQ · NOW I THINK Where can we observe evidence of energy causing change?

We see evidence of energy when things move, make sound, light up,

heat up, or change shape.

Sample answer — student wording may vary.